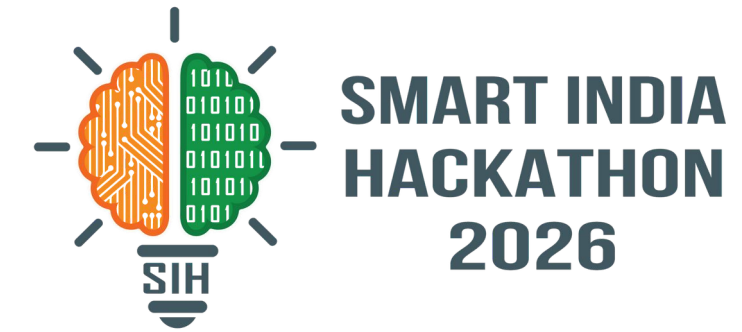
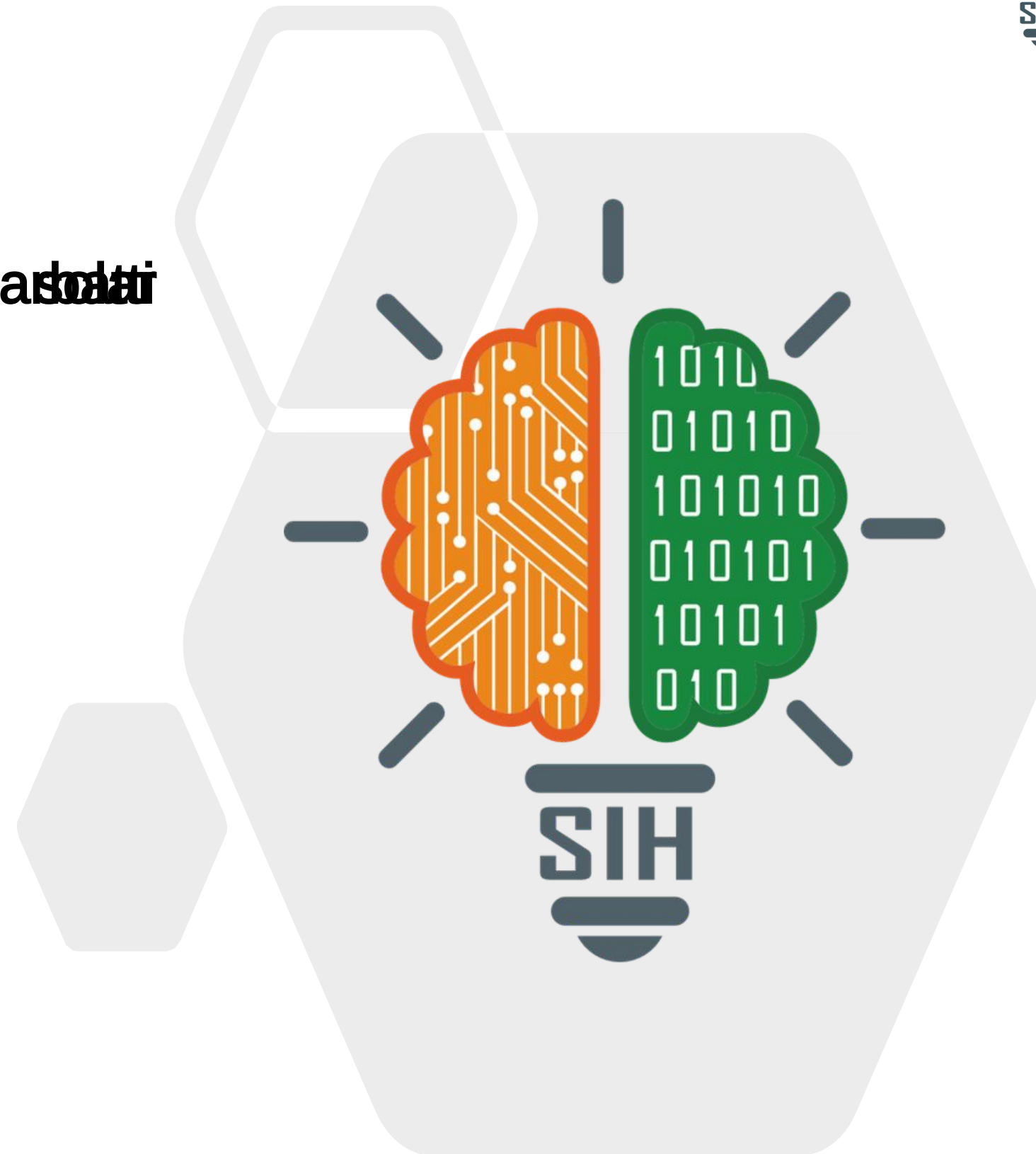


SMART INDIA HACKATHON 2026



- Problem Statement ID –SIH26022
- Problem Statement Title-Smart agasthi
drying ,quality control and
packaging system for rural artisans
- Theme-Smart automation & iot
- PS Category- Hardware
- Team ID-IR2026-1044774
- Team Name –AGNI FORGE



The Problem

Rural women artisans often dry agarbatti using open sunlight or basic drying methods. This creates several problems:



Drying time changes with weather and humidity.



Uneven drying can occur between different parts of a batch.



Over-drying can affect product quality.



Under-drying can cause problems during handling/packaging.



Artisans don't have real-time information about temperature, humidity, drying progress, or energy usage.



Fixed-time drying means the dryer may keep running even after the batch is sufficiently dry.



Solar energy is available but isn't intelligently managed.

Our Solution

AGNI FORGE – an adaptive, solar-assisted drying and intelligent quality-control system designed for small-scale agarbatti manufacturers.



The system continuously observes the drying conditions and adjusts the dryer instead of blindly following a fixed timer.

Key Idea



Solar Powered
Clean & Renewable
Energy



Smart Sensing
Real-time
Monitoring

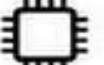
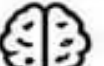
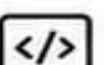


Adaptive Control
Adjusts Fan/Heater/
Ventilation

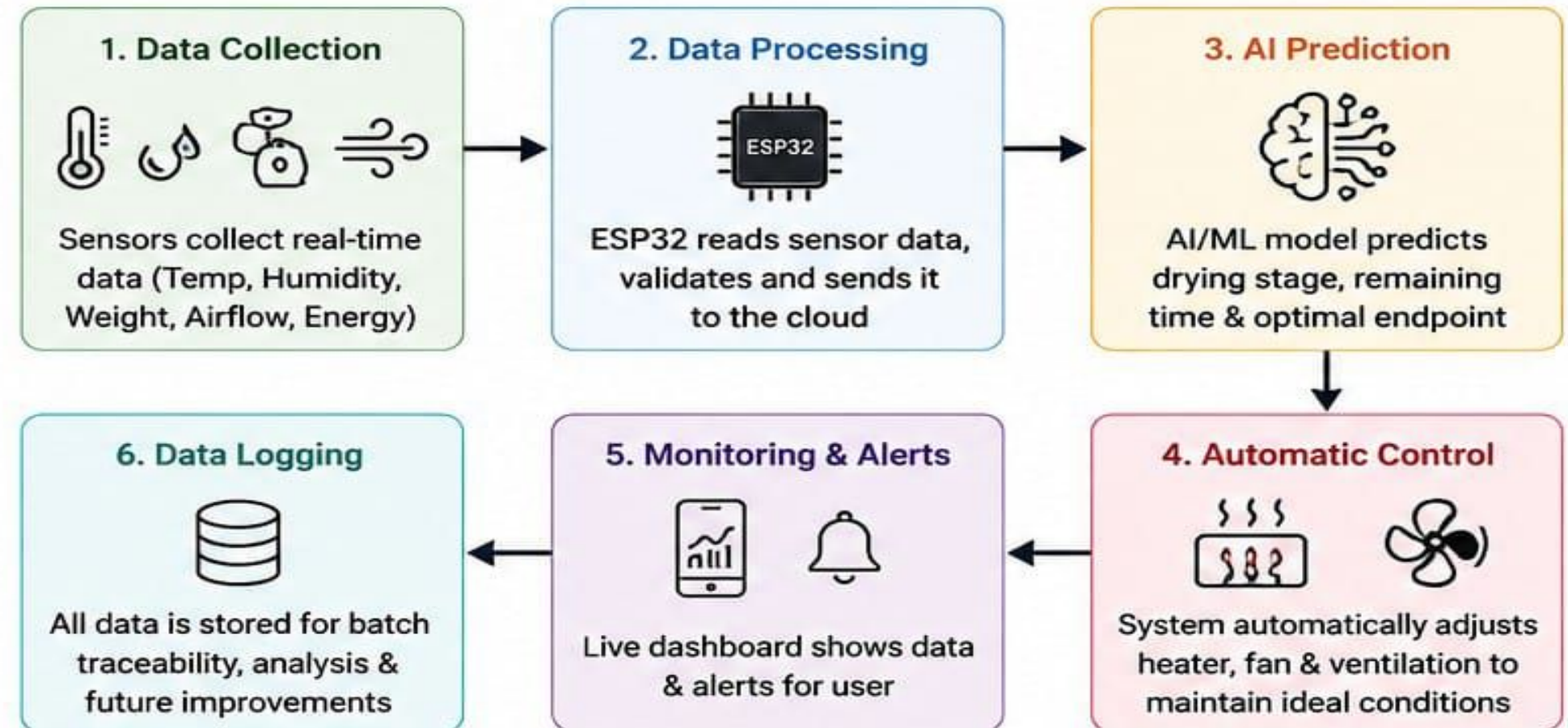


Quality Output
Consistent &
Well Dried Agarbatti

Technologies to be used (e.g. programming languages, frameworks, hardware)

	Controller	: ESP32
	Sensors	: DHT22, DS18B20, Load Cell (HX711), Anemometer, Voltage & Current Sensor
	Actuators	: Heater, DC Fan, Ventilation System
	Power Supply	: Solar Panel + Battery
	IoT Platform	: Blynk / Web Dashboard
	AI/ML Model	: Random Forest / Regression Model (Drying Time & Endpoint Prediction)
	Software	: Embedded C/C++, Python Cloud Database (Firebase/MySQL)

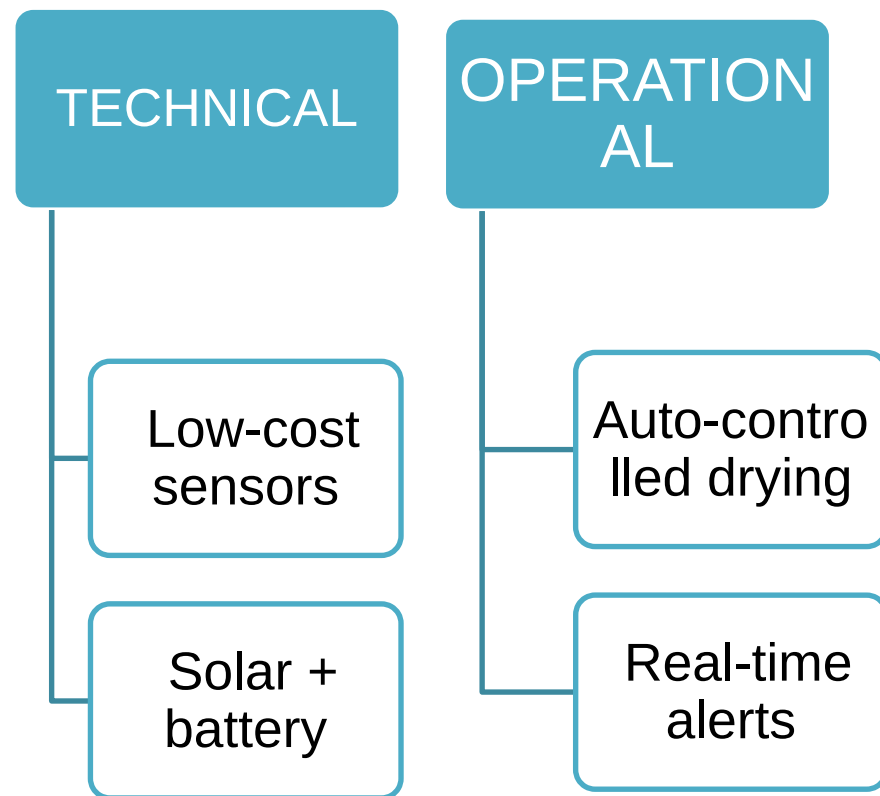
Methodology and process for implementation (Flow Charts/Images/ working prototype)



Working Prototype Overview



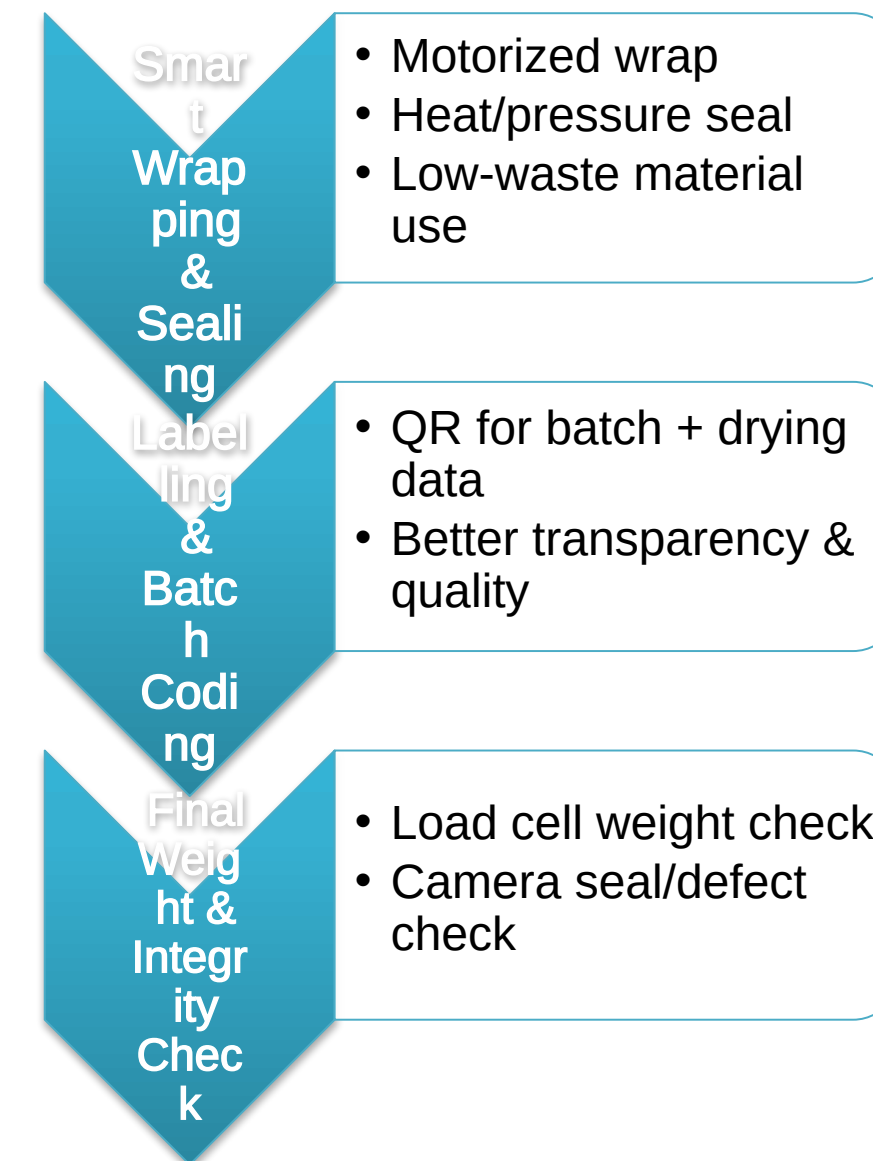
FEASIBILITY



VIABILITY

- Solar drying cuts energy cost drastically
- Expandable from single unit to multi-rack systems
- Aligns with rural livelihood & clean-energy goals

PACKAGING



IMPACT AND BENEFITS

IMPACT



Faster and
consistent drying



Reliable and
affordable for rural
artisans



Higher product
quality



Increased
productivity and
income

BENEFIT



Social benefit



Economic
benefit



Environment
benefit



Decision
making

FUTURE ENHANCEMENT

REFERENCES



Solar-assisted drying reduces dependence on conventional energy sources and improves energy efficiency.



Temperature, humidity and airflow strongly influence drying performance and product quality.



IoT sensing enables real-time monitoring of drying conditions and system parameters



feedback control can automatically regulate fan, heater and ventilation for optimal drying



Batch weight data provides accurate drying progress and helps predict the end point



AGNI FORGE extends these concepts using

batch weight + sensor data + adaptive control

for smart and sustainable drying.



AI Vision

Use computer vision to automatically detect product quality and defects.



Predictive Maintenance

Predict equipment faults and schedule maintenance before failure.



Remote Monitoring

Access real-time data and alerts from anywhere via mobile/web app.



Smart Packaging

Automate packaging with quality check and sealing mechanism.



QR Traceability

Provide batch-wise digital traceability for transparency and trust.



Edge AI & Cloud

Enable faster inference at the edge and long-term data storage and analytics in the cloud.

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